INSURE - Usage-Based Insurance

IOT TELEMATICS & FAIR ACTUARIAL UNDERWRITING PLATFORM

Team: **Techsquad**

Live Demo: **insurevsb.vercel.app**

1. THE PROBLEM & CORE CONCEPT (SIMPLE OVERVIEW)

Traditional motor insurance treats all drivers identically. Whether you drive carefully at 35 km/h or dangerously at 100 km/h with no helmet, both drivers pay the exact same annual premium (₹2,000). **INSURE solves this unfairness: Drive safe, pay less. Drive dangerous, pay more.**

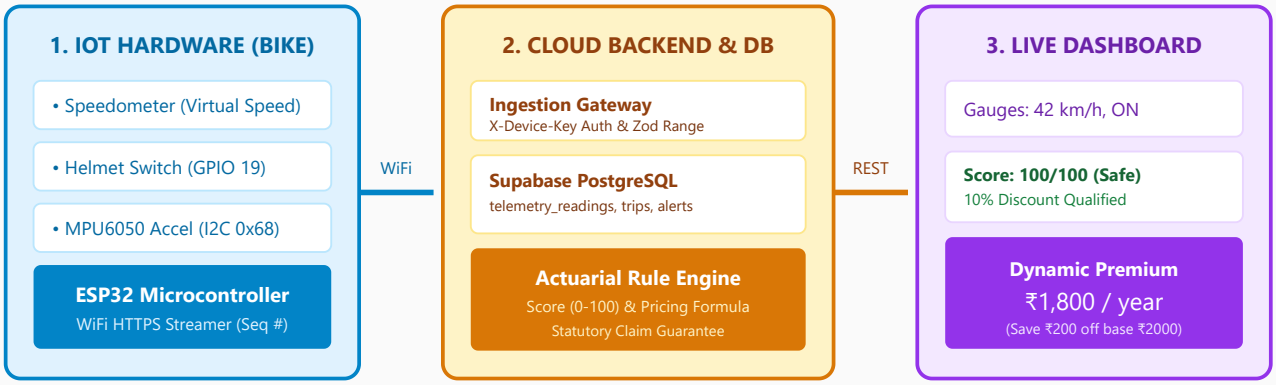
🛵 Safe Commuter (Ravi)

- Drives smoothly at 35–45 km/h
- Always wears helmet (Pin 19 ON)
- Zero alcohol (0.00 ppm)
- **Safety Score: 100/100**
- **10% DISCOUNT → Pays ₹1,800/yr**

🏍️ Risky Commuter (Rahul)

- Speeds above 95 km/h
- Helmet switch OFF (Violation)
- Harsh braking & rapid acceleration
- **Safety Score: < 50/100**
- **+20% SURCHARGE → Pays ₹2,400/yr**

2. END-TO-END SYSTEM ARCHITECTURE DIAGRAM



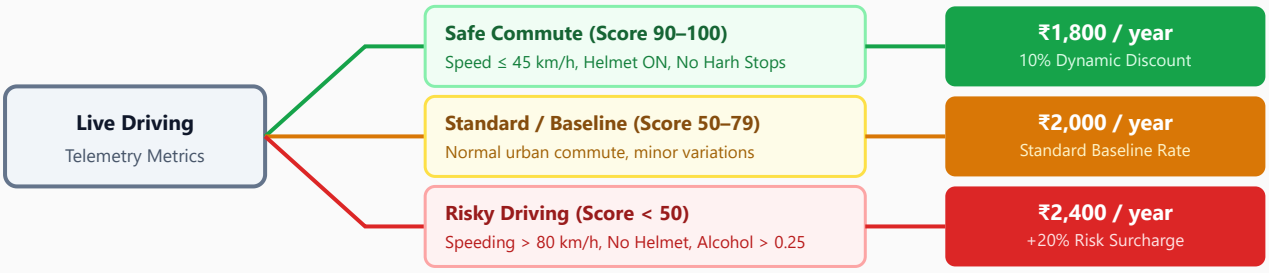
3. THE 5-STAGE STEP-BY-STEP DATA WORKFLOW

1. **Hardware Sampling**: The ESP32 circuit samples sensor pins every 2.5–4.0 seconds (MPU6050, GPIO 19, ADC 34).
2. **Secure Transmission**: Formats JSON packet with sequence number (Seq #10) and posts over WiFi with `X-Device-Key` header.
3. **Gatekeeper Validation**: Cloud gateway verifies device key, ensures speed is realistic (0–300 km/h), deduplicates packet event IDs, and normalizes device timestamps to server clock.
4. **PostgreSQL Storage**: Ingested rows are permanently logged in `telemetry_readings`, linked to active vehicle trip sessions.
5. **Actuarial Scoring & Pricing**: Computes driver risk on 5 dimensions (speeding, braking, accel, cornering, distractions) → adjusts premium to ₹1,800/yr → feeds live UI via 4s polling.

4. ESP32 HARDWARE PIN & SENSOR MAPPING

Sensor Component	ESP32 Pin	Physical Measurement	Telematics Function
MPU-6050 Accelerometer	I2C (SDA 21, SCL 22)	3-Axis Acceleration (m/s^2)	Detects harsh braking (< -4 m/s^2) and impact force (> 5.0g collision)
Helmet Slide Switch	GPIO 19 (Digital In)	HIGH (3.3V) / LOW (0V)	Verifies driver helmet compliance (ON = Safe, OFF = Safety Violation)
Potentiometer (MQ-3)	GPIO 34 (ADC Analog)	0.00 – 0.80 ppm Alcohol	Simulates breathalyzer alcohol detection for zero-tolerance policy
Speedometer Pulse	Firmware Internal	0.0 – 120.0 km/h	Computes highway speeding breaches against speed limit (80 km/h)
Crash Pushbutton	GPIO 18 (Interrupt)	Impulse trigger	Forces high-g crash event (7.42g) to test emergency SOS & provisional claims

5. ACTUARIAL PRICING & FAIRNESS RULE ENGINE



6. STATUTORY FAIRNESS & COMPLIANCE GUARANTEES

- **Separation of Pricing & Claims**: Telematics scores only adjust annual ongoing policy renewal prices. **Legitimate claim payouts can NEVER be reduced, delayed, or denied** based on driving telematics risk scores.
- **Neutral Accident Protocol**: Severe impact forces (> 5.0g) create an unverified provisional incident. Under our fairness charter, driver fault is marked strictly as **UNKNOWN** until formally validated by a certified human investigator.
- **No Zero Penalties for Calibration**: New drivers or short trips under 1 km operate on an approved baseline without unfair algorithmic penalties.